

# ABO Groups, Rh Disease and Transfusion Medicine

Principles and Practice of Osteopathic Medicine I

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Make.  
Heal.

Innovate.  
Reinvent the Future.

# Session Objectives

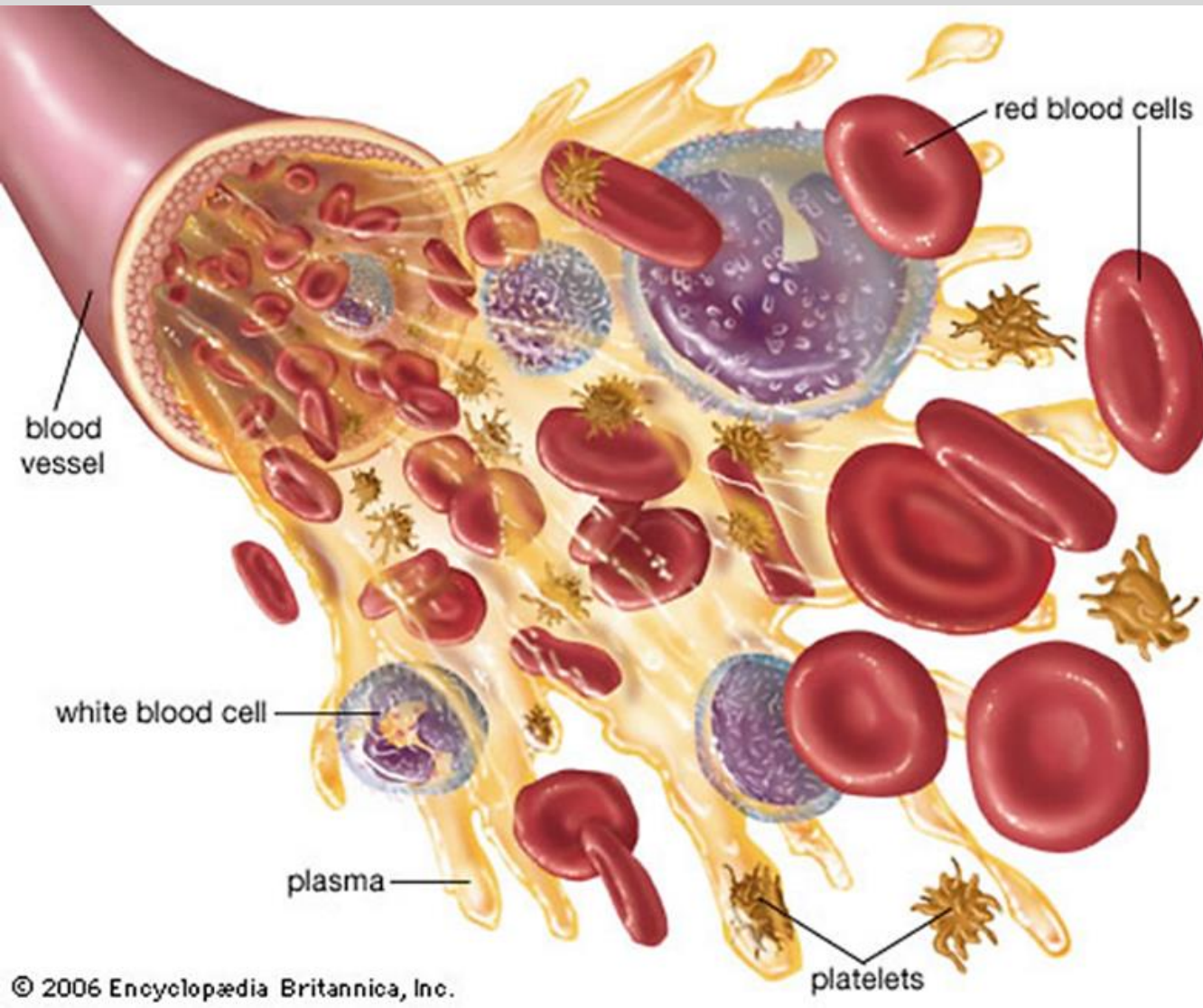
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1. Understand the **immunologic basis** of transfusion medicine, in particular the **ABO** and **Rh** blood group systems
2. Learn about the different **components** of whole blood and the **indication** for transfusing packed red blood cells, platelets, plasma, cryoprecipitate, albumin, IVIG, and factor concentrates
3. Understand and be able to apply the criteria for **appropriate** transfusion practice
4. Know the **alternatives** to transfusions
5. Understand the different types of **transfusion reactions**, their etiology and the evaluation and management of a patient who has experienced a transfusion reaction
6. Know the common **transfusion transmitted diseases** and how donors are screened for these

# Blood

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~60% **plasma** (antigens and antibodies)

~40% formed elements (ex **RBCs**, **WBCs**) - antigens

# Safe transfusion of blood

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- Humans belong to different **blood component groups** – (ex. RBC) surface antigens
  - With blood transfusion, **immune system** reacts with **antigens different from self** → antibodies against antigens they lack
    - For safe transfusions, ensure transfused antigens **match** patient's **blood type**
    - Potential complications/transfusion reactions include life threatening
      - Hemolysis → shock, acute renal failure, DIC
      - Anaphylaxis → from urticaria to shock

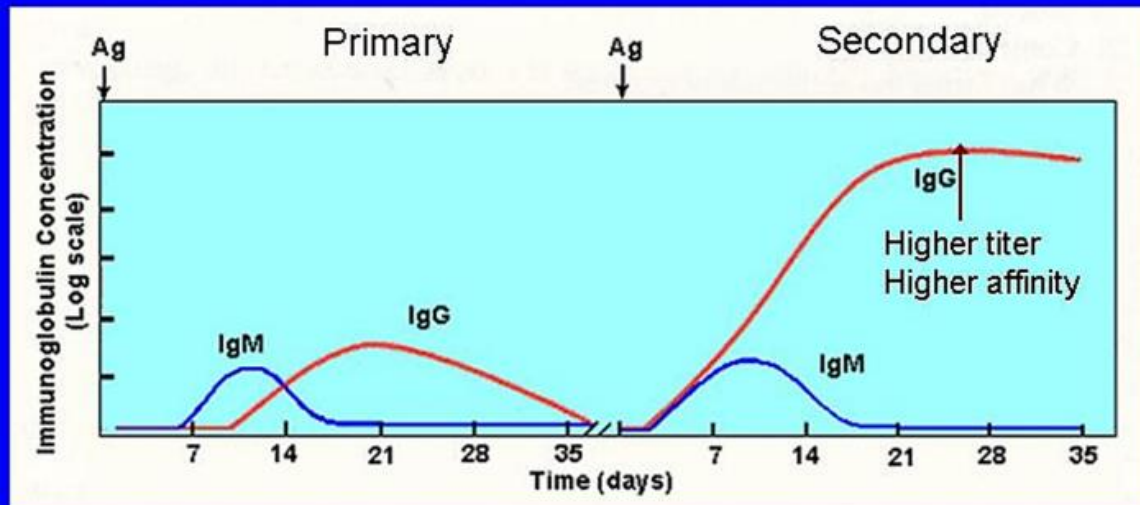
# Transfusions and antibodies

- **Natural: Never exposed by transfusion, injection or pregnancy**
  - Mostly **cold** reacting antibodies (**IgM**, **room temp.**/below body temp., activate **complement**)
  - Most commonly, against RBC ABO/ABH antigens
    - Usually clinically significant when active at 37°C, **anti-ABH, intravascular hemolysis**)
- **Immune: In people exposed to (RBC) antigens**
  - Mostly **warm** reacting antibodies (**IgG** that bind antigen best at **37°C** – body temp.)
  - Most common ones include anti-**Rh**, -Kell, -Duffy, and -Kidd
    - May be hemolytic (usually via **opsonized** RBCs, **extravascular** hemolysis)
  - Require use of anti-human globulin (**Coombs reagent**) for detection

# Cold and warm reacting antibodies

- Note that after antigen (Ag) stimulation is removed, titers fall to usually undetectable levels within days to weeks (memory B cells remain)
- Most **cold** antibodies **clinically insignificant**
  - Common: Anti-Le<sup>a</sup>, -Le<sup>b</sup> and -M, -N, and -P (groups)
  - **ABO** group: **Major exception** about “clinical insignificance”
    - **Acute/immediate hemolytic transfusion reactions** at body temperature
- **Warm** antibodies usually associated with **delayed hemolytic transfusion reactions** and hemolytic disease of the fetus and newborn, **HDFN**

## The Antibody Response



## Properties of the major classes of antibodies in humans

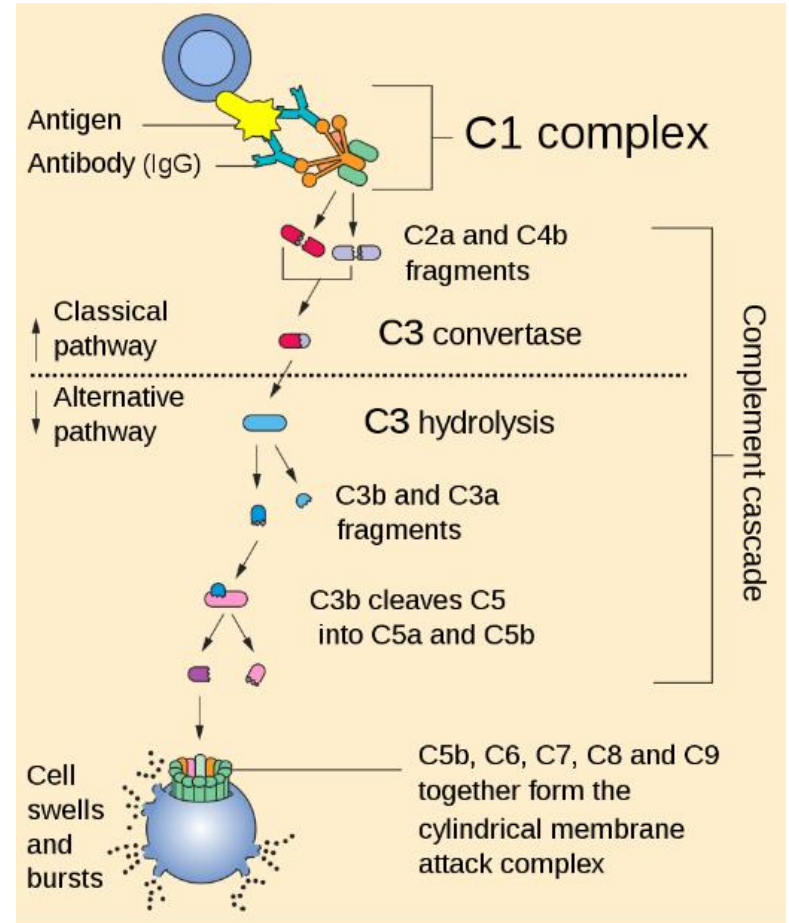
| PROPERTIES                           | CLASS OF ANTIBODY     |                       |                       |                       |                       |
|--------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                      | IgM                   | IgD                   | IgG                   | IgA                   | IgE                   |
| Heavy chains                         | $\mu$                 | $\delta$              | $\gamma$              | $\alpha$              | $\epsilon$            |
| Light chains                         | $\kappa$ or $\lambda$ | $\kappa$ or $\lambda$ | $\kappa$ or $\lambda$ | $\kappa$ or $\lambda$ | $\kappa$ or $\lambda$ |
| Number of four-chain units           | 5                     | 1                     | 1                     | 1 or 2                | 1                     |
| Percentage of total Ig in blood      | 10                    | <1                    | 75                    | 15                    | <1                    |
| Activates complement                 | ++++                  | -                     | ++                    | -                     | -                     |
| Crosses placenta                     | -                     | -                     | +                     | -                     | -                     |
| Binds to macrophages and neutrophils | -                     | -                     | +                     | -                     | -                     |
| Binds to mast cells and basophils    | -                     | -                     | -                     | -                     | +                     |

# RBC antibodies and complement

**Table 5-2 Antibodies Capable of Binding Complement**

| Most            | Some            | Rare            |
|-----------------|-----------------|-----------------|
| ABO             | Xg <sup>a</sup> | D               |
| Le <sup>a</sup> | LKE             | P <sub>1</sub>  |
| Le <sup>b</sup> | Lan             | Lu <sup>a</sup> |
| Jk <sup>a</sup> |                 | Lu <sup>b</sup> |
| Jk <sup>b</sup> |                 | Kell            |
| Sc1             |                 | Fy <sup>a</sup> |
| Co3             |                 | Fy <sup>b</sup> |
| Ge2             |                 | Co <sup>a</sup> |
| Ge3             |                 | Co <sup>b</sup> |
| Ii              |                 | Di <sup>a</sup> |
| P               |                 | S               |
| P1PK            |                 | s               |
| Vel             |                 | Yt <sup>a</sup> |

[https://nyit.ent.sirsi.net/client/en\\_US/default/search/results?qu=Modern+Blood+Banking+and+Transfusion+Practices&te=#](https://nyit.ent.sirsi.net/client/en_US/default/search/results?qu=Modern+Blood+Banking+and+Transfusion+Practices&te=#)



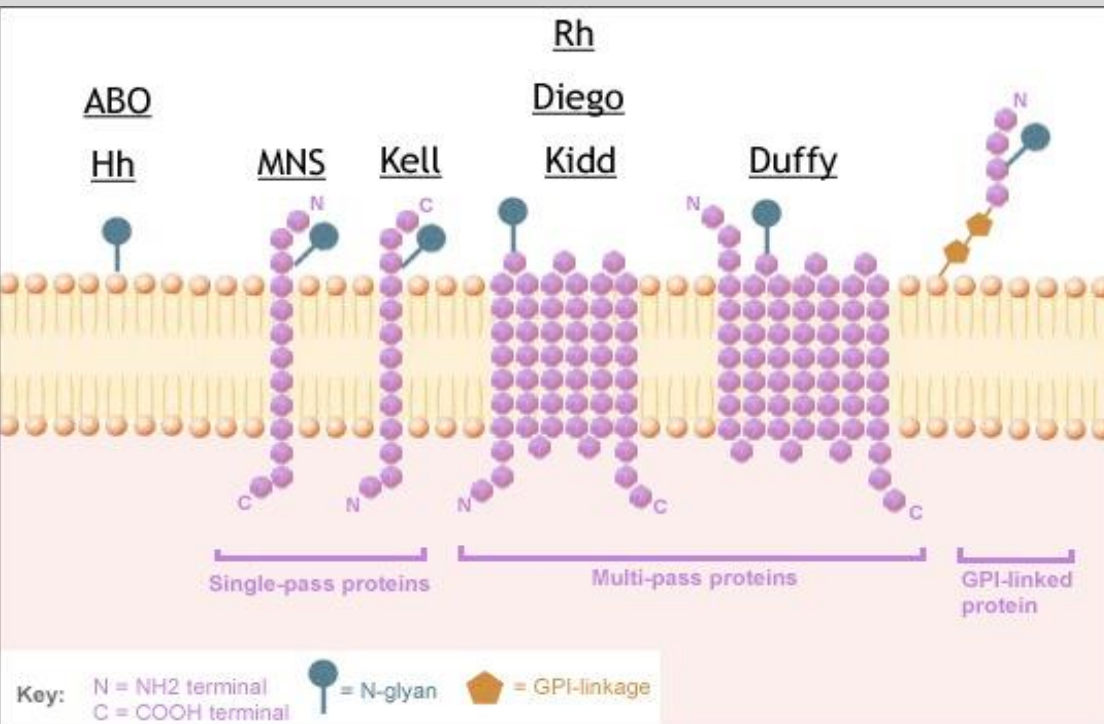
[http://www.niaid.nih.gov/publications/immune/the\\_immune\\_system.pdf](http://www.niaid.nih.gov/publications/immune/the_immune_system.pdf)  
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# Common RBC antigens

- **Extracellular regions** of antigens are antigenic and **various parts of antigens (epitopes)** are recognized by the immune system

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Blood Groups and Red Cell Antigens [Internet].  
<https://www.ncbi.nlm.nih.gov/books/NBK2264/bin/imagemap.jpg>

# Most important red cell blood group systems for transfusion

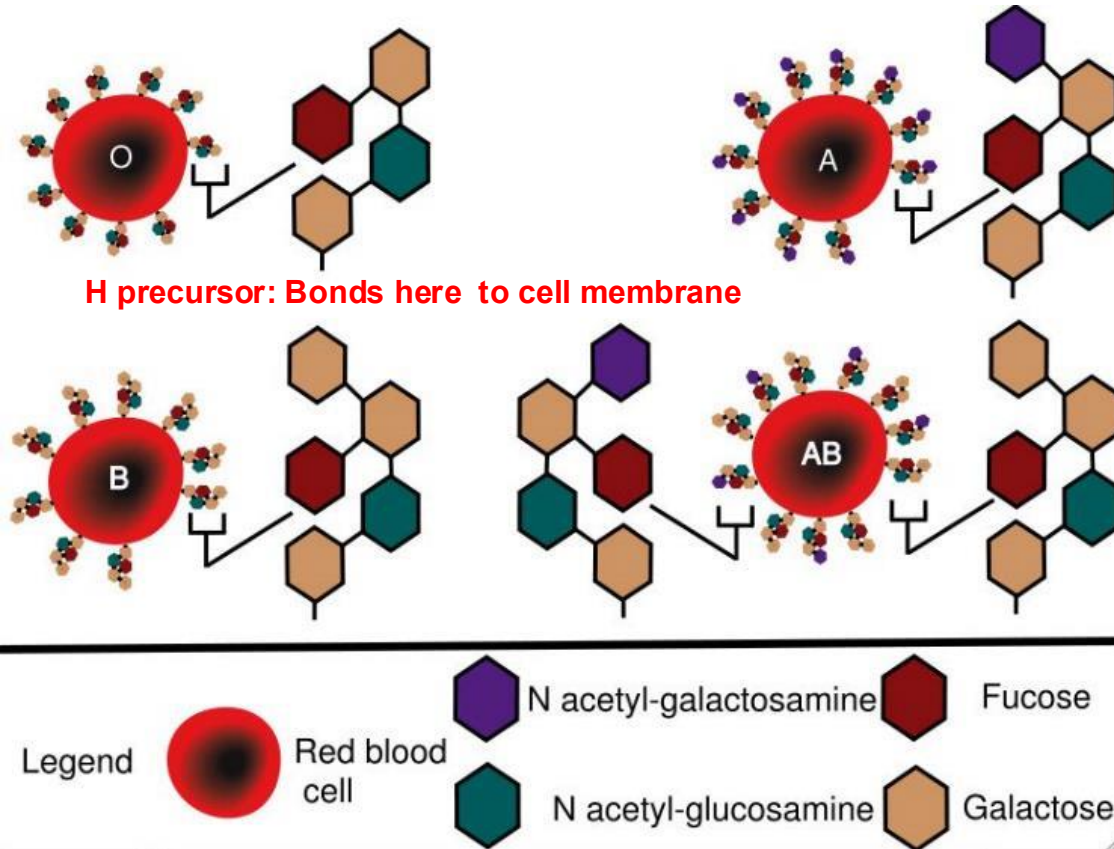
- RBC antigens that are **proteins** (ex. Rh) are **direct** gene products
- RBC antigens that are **carbohydrates** (ex. ABO) are determined/created **indirectly** by the action of enzymes
  - **Glycosyltransferases** add carbohydrates to a basic precursor substance oligosaccharide chain called paragloboside
    - **H gene** transfers a **terminal fucose** to paragloboside → **H antigen/substance**
    - **O gene** does not produce active enzyme → H substance remains unmodified
      - » O is **unmodified H** precursor, without A or B antigens

# Most important blood group systems for transfusion

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- For suitability in blood transfusion, **ABO** and **Rh** determine someone's **blood type/group**
- **Clinical significance** is due to potential to cause hemolytic transfusion reactions:
  - Most importantly **immediate/acute** or **delayed hemolytic transfusion reactions** (**hemolytic disease of the fetus and newborn**)
    - **IHTR/AHTR or DHTR (HDFN)**
  - **ABO** is associated with immediate transfusion reactions
    - **Incompatible ABO type** transfusion → almost **immediate rapid intravascular hemolysis** of donor RBCs
  - **Other RBC antigens** are unexpected/immune and acquired by **alloimmunization** (**prior sensitization** to RBCs):
    - Most significant antigens associated are **Rh** (D, C, c, E, e), Kell (K, k), Kidd (Jka, Jkb), Duffy (Fya, Fyb)
    - Usually **delayed** hemolysis via **extravascular** hemolysis

# ABO blood group system **antigens**

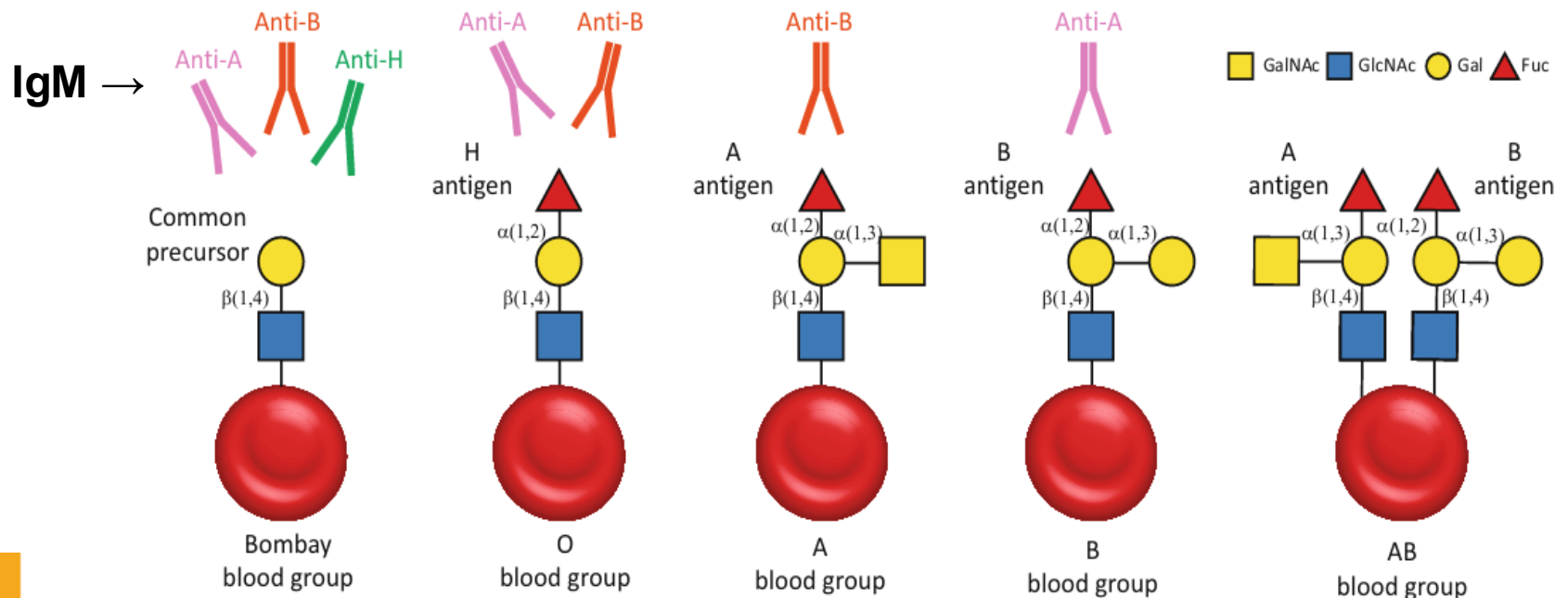


From Wikimedia Commons, the free media repository

- ABO **antigens** are **most immunogenic** of all blood group antigens
- ABO genes produce enzymes that form **complex carbohydrates**
- Antigens vary due to terminal sugar epitopes
  - O (H): fucose
  - A: N-acetyl-galactosamine
  - B: galactose

# ABO blood group system antigens & antibodies

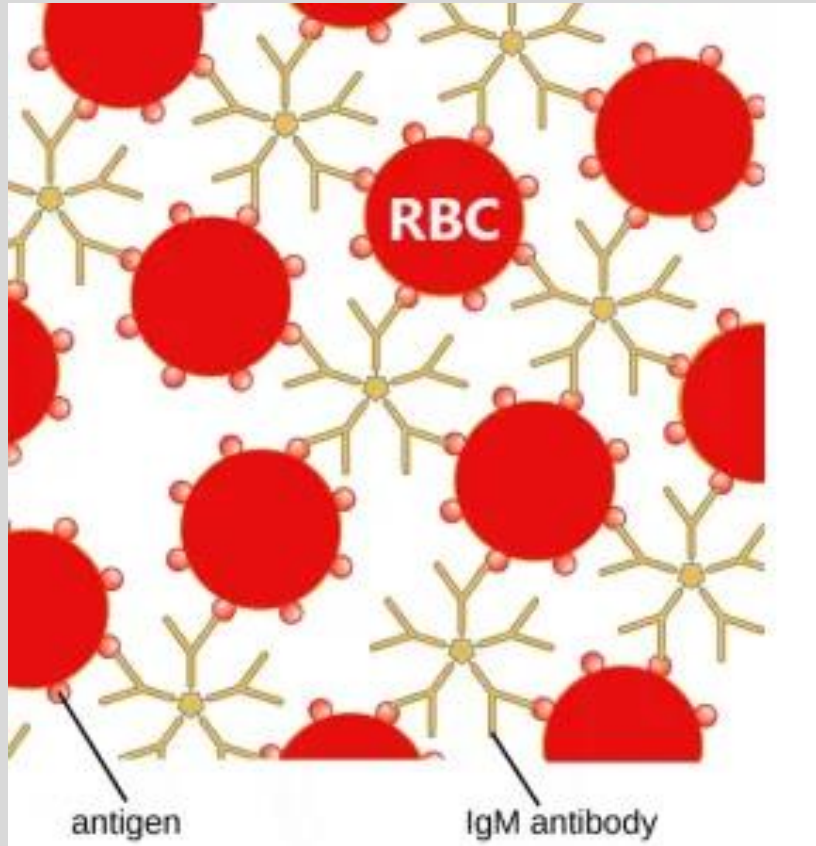
- Blood type O is the most common worldwide, then group A, then B and AB
- Bombay** phenotype is **rare** and due to a **lack** of the **H** antigens
  - Also, **naturally** occurring **anti-A**, **-B** and **-H** may produce intravascular **hemolysis**
  - Problem to transfuse these persons:
    - Essentially, **everyone** has **H substance**, except this phenotype
    - May only receive blood from another person with this phenotype



# Red blood cell ABO antigens & IgM antibodies

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<https://microbeonline.com/coombs-test-types-principle-results/>

- **Agglutinins** are antibodies that cause **agglutination**
- Antigen-antibody complex forms visible **clumping** in testing system
- **Laboratory forward grouping** detects patient's RBC sample **ABO antigens**
  - **Immediate spin** centrifugation at room temperature
  - Checked for **hemagglutination**
- **Laboratory reverse grouping** serves as **confirmation** of RBC forward grouping results
  - Patient's serum (**antibodies**) is added to reagent A and B cells
  - Tested at immediate spin for **hemagglutination**

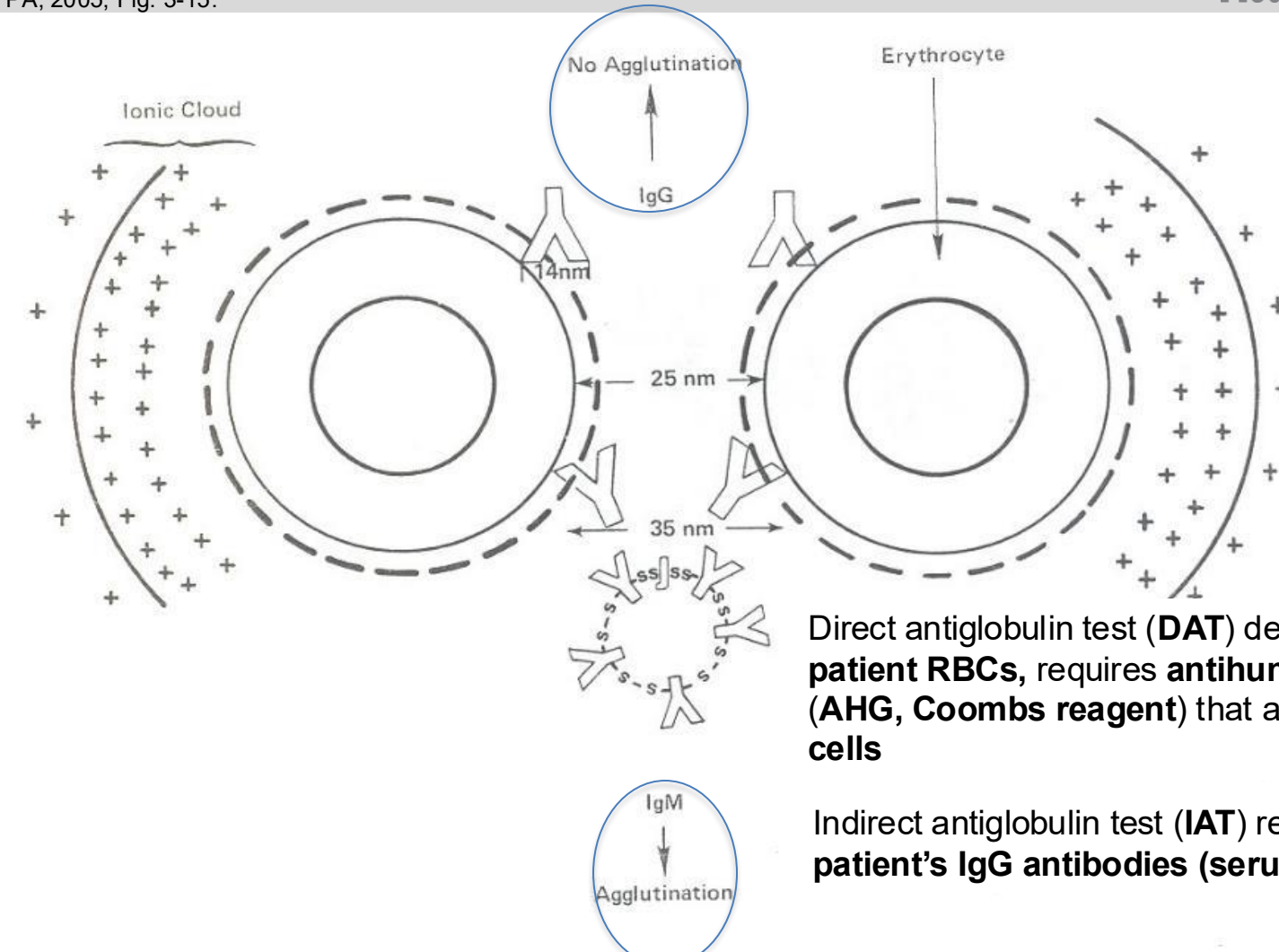
- Typical **blood typing** routinely involves both **ABO** (forward and reverse testing), and **Rh** typing

# RBC ionic cloud diagram: Antibodies (IgG and IgM) relevance to hemagglutination

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Harmening, DM: Modern Blood Banking and Transfusion Practices, 5<sup>th</sup> ed.. F.A. Davis Company, Philadelphia, PA, 2005, Fig. 3-15.



- **DAT and IAT tests detect antibodies** that do not cause direct agglutination, particularly **IgG**, which is too small to hemagglutinate cells itself after centrifugation

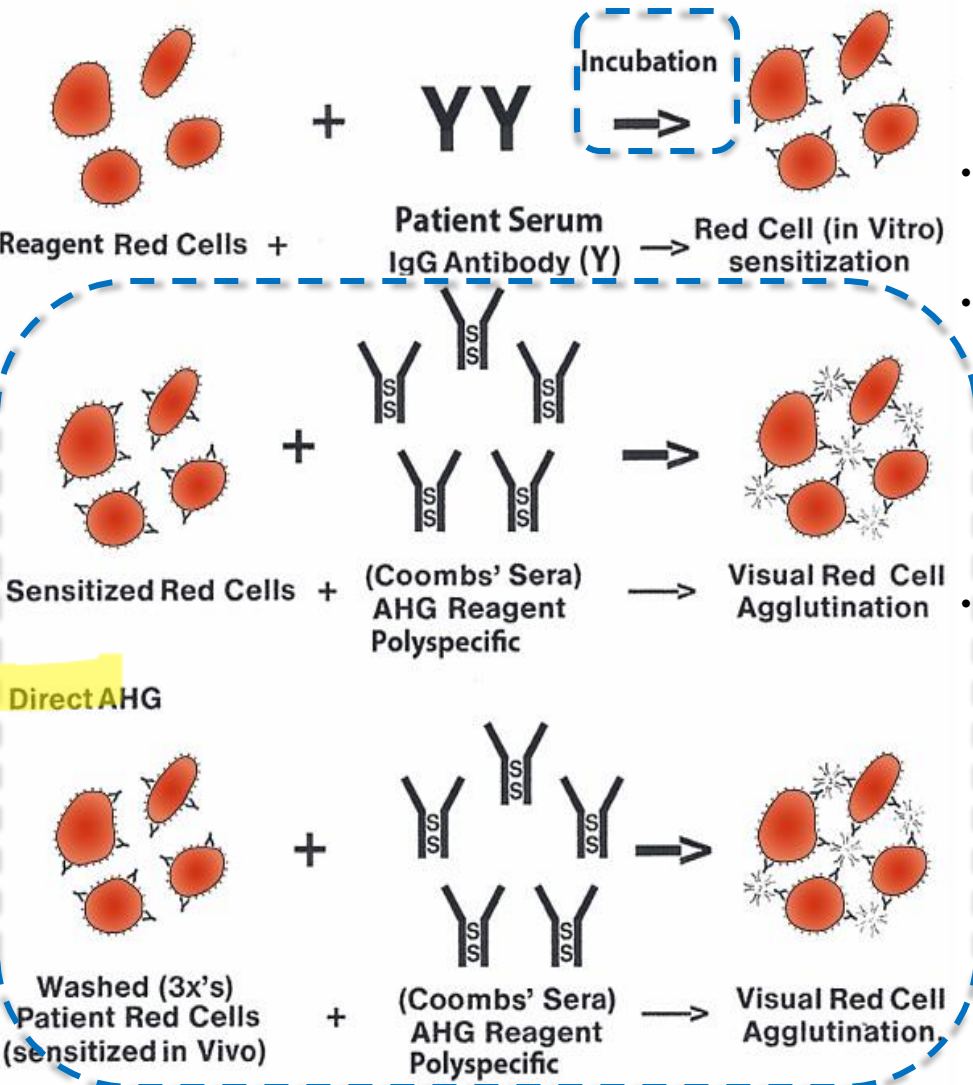
Direct antiglobulin test (**DAT**) detects **IgG/C3** attached to **patient RBCs**, requires **antihuman globulin reagent (AHG, Coombs reagent)** that acts as a bridge to **sensitized cells**

Indirect antiglobulin test (**IAT**) requires **incubating patient's IgG antibodies (serum)** with **reagent cells**

# The Antihuman Globulin Test (AHG)/Coombs test indications:

Harmening, DM: Modern Blood Banking and Transfusion Practices, 6th ed.. F.A. Davis Company, Philadelphia, PA, 2012, Fig. 5-3

## Indirect AHG



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- **DAT** is part of clinical work-ups for HTR and HDFN
  - Detects **in-vivo sensitization of patient's RBCs** with **IgG/C3**
- **IAT** detects **patient's (serum) unexpected antibodies in-vitro** sensitization of reagent RBCs
  - Part of clinical work-ups for HTR & HDFN
  - **Antibody screen:** Part of **type & screen (T&S)**
    - Determines blood type and screens for **patient unexpected antibodies** against reagent screening cells
- Major **crossmatch** tests donor red blood cells against recipient serum: Necessary for all transfusions
  - Immediate Spin: Detects **ABO incompatibility only**
  - **Incubation** (37°C) for significant unexpected IgG antibodies & **antiglobulin** phases after incubation
  - Type & cross (T&C) involves typing & major crossmatch
    - Removes requested blood from inventory/**unavailable** for others

# Rh blood group system

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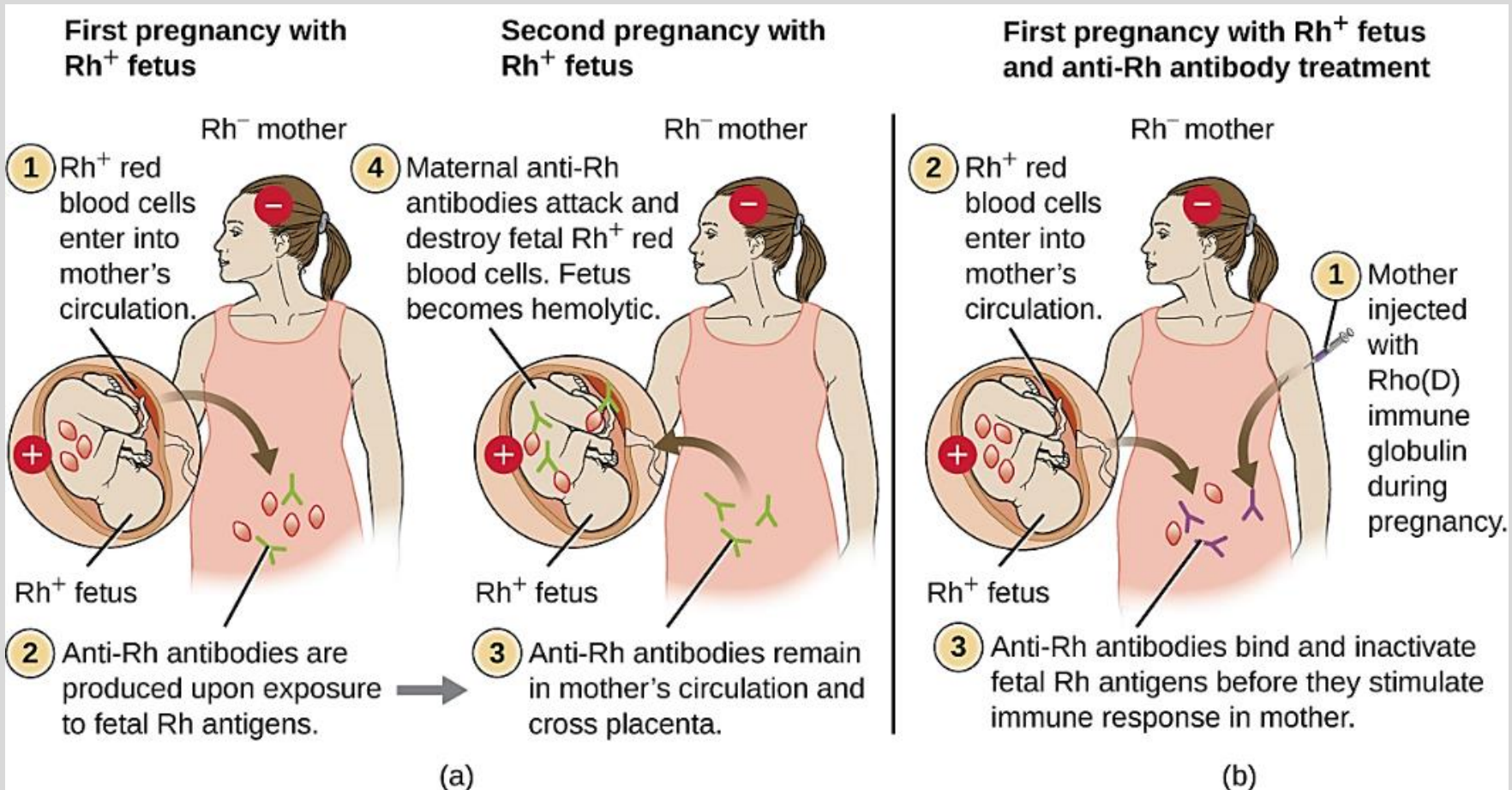
- Clinically, the **most important** blood group **after ABO**
  - Most frequent target of **alloantibodies** caused by transfusions, and in mothers alloimmunized to fetal antigen (**HDFN**)
- The first Rh antigen defined was the D antigen: The term **Rh positive/negative refers to the D phenotype** (A, B, AB, and O, with **+ or – denoting Rh(D)** status)
- But Rh blood group system is comprised of **> 40** individual antigens
  - Most **common Rh antigens** routinely identified in tests are **D (most immunogenic) > c > E > C > e**

# HDFN (aka erythroblastosis fetalis) and Rh immune globulin

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- ~95% of cases by maternal antibodies formed were against **Rh(D)** before use of Rh(D) immune globulin (**RhIG**)



OpenStax Microbiology. Provided by: OpenStax CNX. Located at: <http://cnx.org/contents/e42bd376-624b-4c0f-972f-e0c57998e765@4.2>.

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# Most common transfusion reactions: Estimated frequency

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- Transfusion reaction **~1% overall incidence**, most common:
  - **Allergic** (urticarial) 1-3%
  - **Febrile** Non-hemolytic 1-3%
  - **TACO**, Transfusion-Associated Circulatory Overload 1/200
  - **Delayed** Hemolytic Reactions (>24 hours, extravascular) 1/2,500
  - **Septic** transfusion reaction, (**Bacterial** contamination) 1/5000
  - **TRALI**, Transfusion-Related Acute Lung Injury 1/10,000
  - **Anaphylactic** 1/50,000
  - **Acute/Immediate** Hemolytic Reactions (<24 hours, intravascular) 1/75,000
  - Transfusion-associated graft-versus-host disease (**TA-GVHD**) 1/500,000
- Some **overlapping signs and symptoms** → this may lead to misdiagnoses

# DHTR (extravascular) and AHTR (intravascular) clinical features: Lab evaluation and management

- **DHTRs** often involve **Rh** system
  - ↓**Hgb**, ↑**bilirubin**, new **antibody**
    - Characterized by **hematocrit** that fails to rise, or decreases
    - Development of a positive **DAT**
- Most common cause of **AHTR** is **ABO** incompatibility
  - Rarely, **Kidd**, **Rh**, **Kell**, and **Duffy** antigen incompatibility also been implicated

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- **AHTR** → **Stop blood transfusion**
- Maintain blood **pressure** (intravenous fluid, vasopressors, diuretics) & **airway**
- **Return** blood and transfusion set to lab & order appropriate labs: Check for serum hemolysis
  - Peripheral blood **schistocytes** & **brown serum**
  - Increased **LDH** (due to RBC lysis)
  - Increased total & indirect **bilirubin** (**jaundice**)
  - Decreased **haptoglobin** (overwhelmed)
  - **Hemoglobinuria** (free hgb in the urine, pink urine)
  - **Hemoglobinemia** (free hgb in the blood)
- Monitor renal function (**BUN**, creatinine)
- Repeat compatibility testing (**crossmatch**)
- Repeat for **DAT**, **IAT**

### Delayed Hemolytic Transfusion Reaction

- Onset of symptoms: 5-10 days after RBC transfusion
- S/S: hemolytic anemia, jaundice, fever (can also be asymptomatic)
- Life-threatening complications are rare
- Confirmation: repeat type and screen to detect alloantibody
- Treatment: supportive

### Acute Hemolytic Transfusion Reaction

- Abrupt onset of S/S
- S/S: intravascular hemolysis, hypotension, fevers, AKI, pain at the infusion site, DIC, pink plasma or urine
- Treatment: stop the transfusion!
  - Send blood back to blood bank to check for incompatibility, hemolysis
  - Supportive treatment with IVF, pressors, diuresis

# Febrile nonhemolytic transfusion reactions

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- **FNTR** reactions are due to
  - **Cytokines** released during component storage
  - **Antibodies** in patient's plasma to **donor leukocytes** or **platelets**
- More common in **multiply transfused** patients or **multiparous** women
- **Fever**, often accompanied by **chills**, and sometimes **nausea/vomiting without** evidence of **hemolysis or allergic reaction**
- Reactions usually **more uncomfortable than** clinically **serious**
- Stop and monitor for progression of symptoms, **antipyretics**

# Allergic transfusion reactions

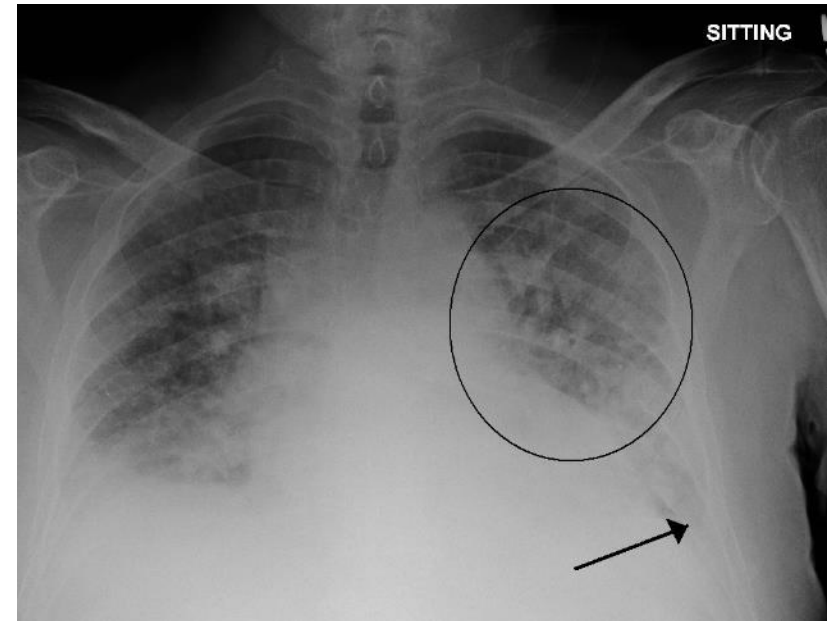
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- Caused by **antibodies** from prior exposure to **donor plasma proteins**
- Most common with **plasma-rich components** like platelets or FFP
- Allergic **manifestations** are **immediate**: **Urticaria** common, **bronchospasm**, **anaphylaxis**
  - **Difficulty breathing** symptoms of anaphylaxis require rapid treatment
- Treatment: **Stop transfusion** and **monitor** for **progression**, **diphenhydramine**, possibly **oxygen**, **hydrocortisone** or **epinephrine** for anaphylaxis

# Transfusion-associated circulatory overload (TACO)

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- **Cardiogenic pulmonary edema** due to **volume overload** (hypervolemia)
- Sx/Sx: Systolic hypertension, dyspnea, hypoxemia, tachycardia, crackles
- **Risk factors**: Diseases that increase blood volume, including **liver, heart, or kidney failure, many transfusions**
- **Managed** similar to **congestive heart failure**



Interstitial and alveolar **pulmonary edema** (circle) with **pleural effusions** on both sides (arrow)

<https://upload.wikimedia.org/wikipedia/commons/1/1c/PulmEdema.PNG>

# Transfusion-related acute lung injury (TRALI)

- **Non-cardiogenic pulmonary edema** → acute **respiratory distress**, tachypnea, dyspnea, hypotension
- Previous transfusion or transplantation can lead to **donor sensitization**
  - **Pre-existing patient pulmonary pathology** causes **neutrophils** to accumulate in pulmonary microvasculature, then **anti-neutrophil antibodies** are transfused and react w/ **recipient leukocytes**
  - Ex. **Multiparous** female donors → **higher risk** of TRALI for recipients
- Typically products containing plasma
- Treatment is supportive



TRALI before (left) and after (right) treatment

[https://upload.wikimedia.org/wikipedia/commons/7/73/Transfusion-related\\_acute\\_lung\\_injury\\_chest\\_X-ray.gif](https://upload.wikimedia.org/wikipedia/commons/7/73/Transfusion-related_acute_lung_injury_chest_X-ray.gif)

# Transfusion-associated graft-versus-host disease, TA-GVHD

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- **Immuno-incompetent** host
  - Donor T lymphocytes **engraft** and **attack host** w/ **pancytopenia** and **multiorgan failure** (skin, liver, gut)
- Usually **fatal** (>90% of cases)
- Most common causes of death - **infections** and **hemorrhages**
- **Prevent** by **gamma irradiation**

# Bacterial contamination (septic transfusion reaction)

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- Incidence **higher** with **platelets** than with **RBC**
  - **Gram-negative** bacteria (ex. *Yersinia* and *Pseudomonas*) - **RBCs**
  - **Gram positive** organisms (ex. *Staphylococcus*) - **platelets**
- **Fever, rigors, hypotension (septic shock), death**
- **Stop** transfusion, **antibiotics** and **culture**

# Blood product collection and transfusion is heavily regulated

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- Among other criteria, donors must **generally** be in **good health** and feeling well
- At least 17 years old in most states
- At least 110 pounds
- Not donated blood in last 56 days
- **Generally** temporarily **excluded**: Recent **infection**, fever, **travel** (endemic disease), **drugs** (ex. teratogens), **illicit drug** use

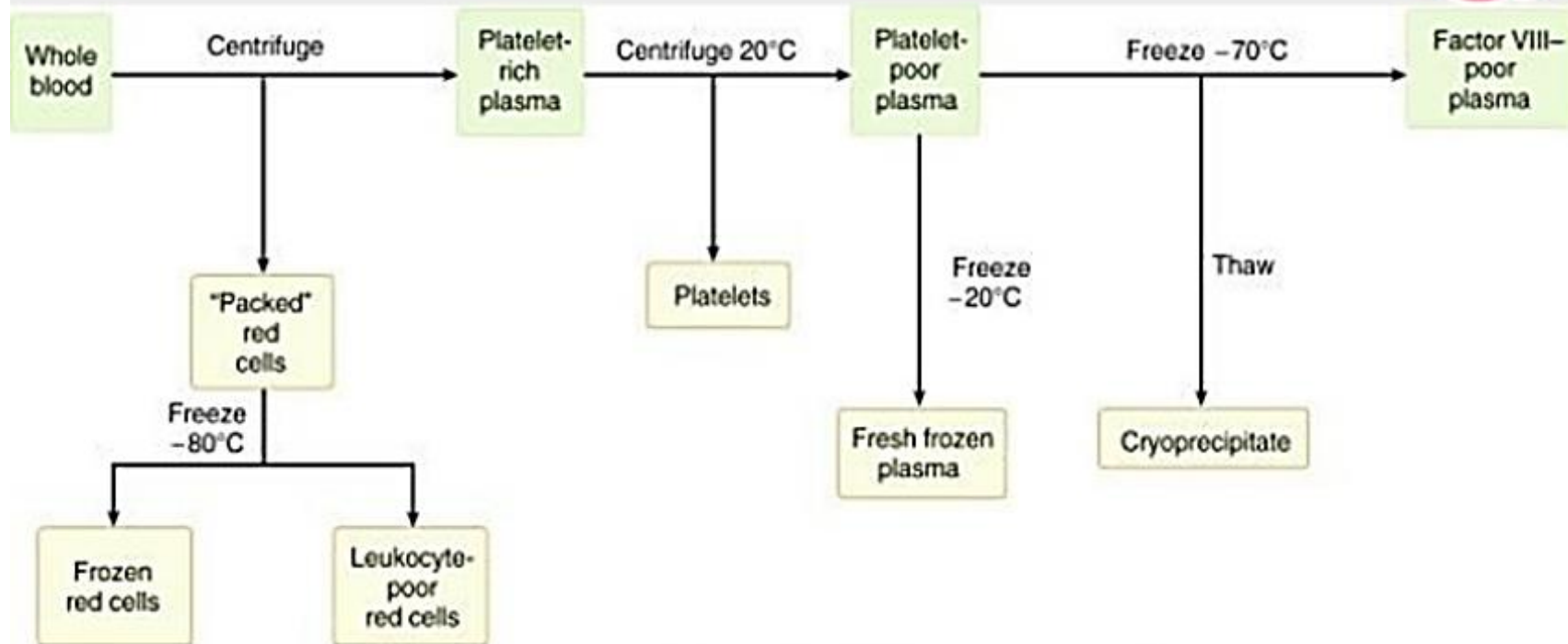
# Estimated frequency of TTD

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- Donor **Screening** Tests

- Hepatitis B: HBsAg, Anti-HBc, HBV DNA
- Hepatitis C: Anti-HCV, HCV DNA
- HIV: Anti-HIV 1/2, HIV-1 RNA
- HTLV: Anti-HTLV I/II
- Syphilis: RPR or IgG to T. pallidum
- West Nile: WNV RNA
- Chagas: IgG to T. cruzi
- Babesia antibodies
- Zika virus RNA
- Hepatitis B <1/800,000 transfusions
- Hepatitis C <1/1,000,000
- HIV <1/1,000,000
- HTLV I and II <1/250,000
- Syphilis No cases >40 yrs
- T.Cruzi ~1:200,000 endemic areas with screening, ~20 cases in US past 20 years
- Babesia <1/100,000 transfusions 2011-2015
- West Nile one per 35 million 2003-2016 donations
- Zika Virus No cases 2016-2019, few cases world-wide

# Transfusion therapy principles: Component therapy



Scheme for separation of whole blood for component therapy.



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- **Whole Blood:** 450-500 ml, 60 ml of **citrate** anticoagulant/preservative solution
  - Citrate **toxicity** (**hypocalcemia**)
  - 21-35 days shelf life, **refrigerated**
  - **Uncommon** use today, except trauma/massive transfusions
- **Plasma, platelets** removed as components
  - Packed red blood cells (**pRBCs**) mixed w/ **additive** (ex. Adsol) - (**lower volume** an advantage)
    - Storage period extended (42 days), **refrigerated**

# Leukocytes and leukoreduction

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- Leukocytes are **contaminants** of cellular components
- Adverse consequences: **Immunologically-mediated** most **common**), also infectious disease transmission
  - **FNHTRs**, **platelet refractoriness**, transplant **rejection**, **GVHD**, immunosuppression
  - Transmission/reactivation of **viral diseases**
- **Leukoreduction**: Filters (arrow) remove leukocytes
  - Prevents **FNHTRs**, **HLA alloimmunization**, **CMV**



# Leukocyte transfusions & blood product irradiation

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- **Neutrophils** occasionally transfused: **Severely neutropenic** or with **infections unresponsive to antibiotics** – Risks may outweigh benefits
- **Immuno-incompetent patients: Gamma or X-ray** irradiation inactivates contaminating T lymphocytes in blood/tissue transplants preventing **TA-GVHD**
  - Indications include,
    - **Premature, low-birthweight infants**
    - **Congenital immune deficiencies**
    - **Hematologic malignancies** and solid tumors
    - Peripheral blood **stem cell or bone marrow transplantation**
    - Components that are **HLA matched**, or **directed donations**
  - **Granulocyte components** should be irradiated and transfused with minimum delay

# Washed blood components & frozen red blood cells

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- Severe/recurrent allergic transfusion reactions: Red cells and platelets can be **washed** using isotonic **saline** to **remove plasma proteins**
- Major indication for **freezing** red cells is to collect **rare donor units**

# Appropriate RBC transfusion indications

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- When **alternative** treatment is **not available**, or **clinical manifestations** in patient demand treatment
  - Levels  **$\leq 7$  gm/dL** currently a guideline for **most patients**
  - Patients with **active** coronary artery disease often transfused at higher hemoglobin levels ( **$\leq 8$  gm/dL**)
- Response to transfusion varies: One red cell unit expected to increase **hemoglobin** by **1 gm/dL** or **hematocrit** by **3%**

# Alternatives to transfusions of donor RBCs

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- Treat the **underlying cause** of anemia
- **Erythropoietin**
- **Autologous** transfusions for mainly elective surgical procedures: Collection and reinfusion of a patient's own blood
  - Pre-deposit **self-transfusion**
  - Acute normovolemic **hemodilution**
  - **Cell savers**

# Perioperative transfusion

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- **Most** surgical procedures do **not require blood** transfusion, however
  - Most w/ HGB <7 gm/dL will have transfusion
  - > 25% estimated blood loss → pRBC transfusion w/ crystalloid resuscitation
- **T&S** prudent to obtain if blood products anticipated prior to surgery
- **T&C** for surgical procedures with a low likelihood of transfusion wastes time and resources

# Massive bleeding and massive transfusion protocol (MTP)

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- **Massive bleeding** has various definitions including replacement of 1 entire blood volume within 24 hours
  - MTP describes the process of **management of blood transfusion** requirements in major bleeding episodes
    - **Locally agreed and specific guidelines:** Clinical, laboratory, blood bank and logistic responses

# MTP concept

TABLE

## Massive transfusion protocol in obstetrics

|         | PRBCs                                         | FFP | Platelets | Cryoprecipitate |
|---------|-----------------------------------------------|-----|-----------|-----------------|
| Round 1 | 6 U                                           | 6 U | 6 U       | 10 U            |
| Round 2 | 6 U                                           | 6 U | 6 U       | 10 U            |
| Round 3 | Tranexamic acid 1 g intravenously over 10 min |     |           |                 |
| Round 4 | 6 U                                           | 6 U | 6 U       |                 |

Consider activating the protocol when hemorrhage is expected to be massive (anticipated need to replace 50% or more of blood volume within 2 hours), bleeding continues after the transfusion of 4 U of packed red blood cells within a short period of time (1–2 hours), or systolic blood pressure below 90 mm Hg and heart rate is above 120 beats per minute in the presence of uncontrolled bleeding. Once activated, blood bank personnel will continue preparing blood products until the surgical team inactivates the protocol. After round 4, if not inactivated, the protocol will start again from round 1.

FFP, fresh-frozen plasma; PRBC, packed red blood cell; Adapted from Pacheco et al.<sup>4</sup>

Pacheco. Massive transfusion protocols in obstetrics. *Am J Obstet Gynecol* 2016.

- **Activated** by a clinician in **response** to massive/severe bleeding
- **Balanced transfusions** (including pRBCs and clotting factors) used in efforts to **avoid dilutional coagulopathy**

# Platelet Transfusion

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- Normal platelet **lifespan 5-10 days** stored at **room temperature** with gentle agitation, usually transfused in 5 days
- Indications: **Thrombocytopenia**
  - **<10,000** / $\mu$ L to avoid spontaneous hemorrhage
  - **< 50,000** with bleeding or surgery
  - **<100,000** with CNS bleed/surgery
  - Also, bleeding or invasive procedures in patient with platelet **dysfunction**

# Platelet alloantigens

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- Two clinically significant ( may be severe thrombocytopenia) but uncommon **syndromes**
  - Due to sensitization to platelet specific alloantigens, most commonly **HPA-1a**
  - **Neonatal** alloimmune thrombocytopenia (**NAIT**)
    - Placental transfer of **maternal antibody**
  - Post transfusion purpura (**PTP**): Most often, previously **non-transfused multiparous women**

# Transfusion of plasma and plasma derivatives

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- FFP indicated for bleeding with **multiple coagulation factor deficiency**:
  - **Liver** disease
  - **DIC**
  - Fluid replacement during **plasma exchange**
    - **Rapid reversal** of **warfarin** effect, if factor concentrates unavailable
- Liquid plasma (never frozen), used for **rapid, immediate transfusion** in MTPs, trauma/hemorrhage until FFP thawed
- Plasma components:
  - Cryoprecipitate
  - Albumin
  - Immunoglobulins
  - Factor concentrates

# Cryoprecipitate

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- Indicated for **fibrinogen deficiency**, **DIC**, **uremic** bleeding, **vWD** (von Willebrand's Disease) - Contains
  - $\geq 80$ -100 units **Factor VIII**
  - $\geq 150$  mg **fibrinogen**
  - 40-70% of the von Willebrand Factor (**vWF**)
  - **Factor XIII**
  - **Fibronectin**

# Albumin

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- ~60% of protein in plasma
- Maintains plasma **colloid oncotic pressure**
- Indications
  - **Nephrotic** syndrome **resistant to diuretics**
  - Volume/fluid replacement in **plasmapheresis**
  - Large volume **paracentesis**

# Immunoglobulins (IVIG, intravenous immune globulins)

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- IVIG prepared by **fractionation** of pools of **human plasma**
- Indications
  - **Primary immunodeficiency** syndromes
  - **Acquired immunodeficiency** (e.g. Chronic lymphocytic leukemia (CLL))
  - Some **autoimmune diseases** (e.g. Autoimmune hemolytic anemia, Immune thrombocytopenia)

# Hemophilia **A**- Factor **VIII** & Hemophilia **B**- Factor **IX** deficiency, von Willebrand Disease

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- Hemophilia A
  - **Recombinant Factor VIII** (aka, antihemophilic factor)
  - **Plasma-derived Factor VIII concentrates**
  - **Recombinant factor VIIa**
  - **DDAVP** (desmopressin)
  - Cryoprecipitate
- Hemophilia B
  - **Recombinant Factor IX**
  - **Plasma-derived Factor IX concentrates**
  - **Factor VIIa**
- von Willebrand Disease
  - **Humate P**
  - **DDAVP**
  - Cryoprecipitate if factor concentrates unavailable

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THANK YOU!

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